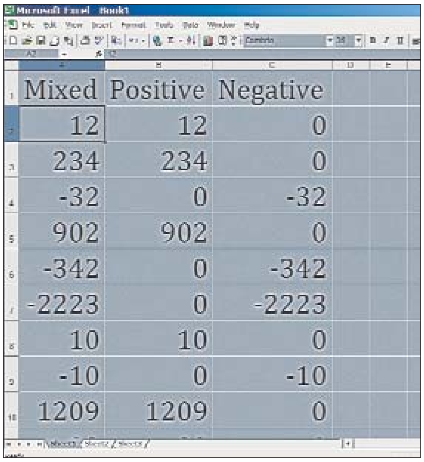
and positive values, you might sometimes want to separate them into columns based on their sign. There are two ways to do this, the first of which is described in this tip.



Mixed	Positive	Negative
12	12	0
234	234	0
-32	0	-32
902	902	0
-342	0	-342
-2223	0	-2223
10	10	0
-10	0	-10
1209	1209	0

Differentiating between positives and negatives

Use a formula in the columns to the right of the column with mixed values. If the mixed column is A, place this formula in the cells of column B:

=IF(A2>0,A2,0)

The result of this will be that column B will only contain values greater than zero. Then, in column C, use this formula:

=IF(A2<0,A2,0)

Column C will only contain values less than zero. So what you have is two new columns of the same length as the original column. Column B is the same as column A with negative values replaced by zero, and column C is the same as column A with positive values showing up as zero.

Numbers With Signs #2

Starting from the above tip, say you want columns that only contain negative or positive values, with no zeroes. You'll need to use Excel's filtering capabilities. Say the mixed values are in column A, with the column heading in A1. You'll need to follow these steps:

In cell E2, place the formula "**>0**". Select any cell in the mixed values of column A. Choose **Data > Filter**, then choose Advanced Filter from the resulting sub-menu. Excel will display the Advanced Filter dialog box. The List Range should already be filled in, representing the range of mixed values in Column A.

Now, make sure the "Copy to Another Location" radio button is selected. Select the Criteria Range box, and with the mouse, select cells **E1:E2**. Select the "Copy To" box and click once in cell **B1**. Then click OK. Column B now contains cells that are greater than zero.

In E2, place the formula "**<0**". Then, once again choose **Data > Filter**, and choose Advanced Filter from the sub-menu that comes up. Excel will display the Advanced Filter dialog box. The settings in the dialog box should be the same as the last time you used them. Select everything in the "Copy To" box, and then click once in cell **C1**. Then click OK. Column C will now contain cells less than zero. Of course, you can now delete the cells at **E1:E2**.

What If...

Suppose you want to calculate how much money you will need to deposit in a savings account for each of 12 months in order to have Rs 10,000 in it at the end of one year. You know the interest rate, but you're not sure of the amount you need to put in. You can use a future value calculator for this (FV). In cell A4 is your deposit amount which is unknown right now. Say cell **A5** contains your interest rate of 5 per cent, and cell **A6** is the number of months (12). In cell **A7**, you can enter the formula:

= -FV(A5,A6,A4)

If you wanted to determine how much you needed to deposit in the account each month, you could repeatedly change the deposit value (A4) until you got close to the desired goal. Here's where the goal seeking tool comes in. Here are the steps:

First select the Future Value cell (the one at **A7**). Choose Goal Seek from the Tools menu. This displays the Goal Seek dialog box. The Set Cell field will have already been set to **A7**. Now move the insertion point to the "To Value" field and change it to 10000. This is the amount you want to have at the end of two years. Then move the insertion point to the "By Changing Cell" field and click on cell **A4** (the cell that will contain your regular deposit amounts). Click OK. Excel tells you it has found a solution. Click OK again. Your worksheet will reflect the solution.

Bet You Didn't Know

A Shortcut For Viewing Formulas

You probably know how to display the formula contents of cells rather than the results: **Tools > Options**, and then in the View tab, you check the Formulas box.

A faster way to do this is to press **[Ctrl] + [`]**. The key is the "backwards apostrophe," just to the left of the **[1]** key. This shortcut toggles, meaning you can use it to switch between the display of formulas and results.

Evaluating Formulas

How exactly does Excel arrive at a particular result? This can be hard to determine, particularly if the formula is complex. Thankfully, Excel provides a tool to help figure out what is going on when it evaluates a formula. To access this tool, first select the cell containing the formula you want to evaluate. Choose **Tools > Formula Auditing**. Excel will display a sub-menu. From the sub-menu, choose Evaluate Formula. Excel will display the Evaluate Formula dialog box.

The full formula from the cell will be shown with part of it underlined. This underlined area represents the part of the formula that will next be evaluated, and this allows you to see what intermediate steps Excel follows in arriving at a result. Every time you click the Evaluate button, Excel replaces the underlined portion of the formula with a result.

Nothing you do with the formula evaluator affects the formula itself. When you're done using the Formula Evaluator, click Close.

Counting Non-Blank Cells

You might already know that you can use the COUNTBLANK function to return the number of blank cells in a range. If you want to count the number of non-blank cells in the range, you could use:

=COUNTA(A1:A10)

The problem here is that this doesn't return the complementary value to what **COUNTBLANK** returns. This is because **COUNTBLANK** and **COUNTA** treat formulas differently: **COUNTBLANK** includes, as blank, formulas that return a blank value. **COUNTA** does not consider such cells blank (even though a blank is returned), so it includes them in its count.

If you define non-blank cells to be those that are not returned by the **COUNTBLANK** function, you'll need to use the following formula:

=(ROWS(A1:B10)*COLUMNS(A1:A10))-COUNTBLANK(A1:A10)

How this works is pretty straightforward, so we'll refrain from explaining it further. ■